Assignment

Paper: *Computer Fundamentals and Programming with C*

Paper Code: *CMSDSC101P* ***Semester:*** *I*

Based on WBSU Guidelines for B.Sc. Computer Science (Hons.) Students.

Prepared for the Students of Acharya Prafulla Chandra College, Department of Computer Science

Basic Numeric and String Processing

1. Write a program to compute the sum and product of digits of an integer using user-defined functions.
2. Write a program to reverse a non-negative integer using a function.
3. Write a program to compute the sum of the first *n* terms of the series using a function: S=1−2+3−4+5−6+…
4. Write a function to check whether a number is prime or not. Use the same function to generate all prime numbers less than 100.
5. Write a function to check whether a given string is a palindrome. Use this function to determine whether an entered string is Palindrome.
6. Write a program using a function to compute and display all factors of a given number.

Macro, Array, and Command-Line Programming

7. Write a program to swap two numbers using a macro (#define).
8. Write a program that counts the number of occurrences of each alphabet (A-Z, a-z) in the text entered using Command-Line Arguments.

Array and Pointer-Based Problem Solving

9. Write a program to swap two numbers using pointers (user-defined function).
10. Write a program in which a function receives addresses of two variables and alters their contents.
11. Write a program that takes the radius of a circle as input, passes it to a function that computes area and circumference, and displays results in main().

Dynamic Memory Allocation

12. Write a program to find the sum of n elements entered by the user. Use dynamic memory allocation (malloc() or calloc()).

13. Write a function that reverses the elements of an array in place, using only a single pointer argument, and return void.

Menu-Driven String Operations (Use Functions Only)

14. Write a menu-driven program to perform the following string operations:
    a. Show address of each character
    b. Concatenate two strings without using strcat()
    c. Concatenate two strings using strcat()
    d. Compare two strings
    e. Find string length using pointers
    f. Convert lowercase to uppercase
    g. Convert uppercase to lowercase
    h. Count number of vowels
    i. Reverse the string

Array and Sorting Applications

15. Write a program to merge two sorted integer arrays to form a single sorted array.

16. Write a program that reads 10 integers into an array (using pointers), and prints the array in ascending and descending order.

Loop, Function & Recursion Applications

17. Write a program to display the Fibonacci series
    (i) using recursion
    (ii) using iteration

18. Write a program to calculate the factorial of a number
    (i) using recursion
    (ii) using iteration

19. Write a program to calculate the GCD of two numbers
    (i) using recursion
    (ii) without recursion

Matrix Handling using Functions

20. Write a menu-driven program to perform the following matrix operations using 2-D arrays and functions:
    a. Sum
    b. Difference
    c. Product
    d. Transpose

File Handling Applications

21. Write a program to copy the contents of a text file to another file, after removing all white spaces (spaces, tabs, newlines).

Structure-Based Programming

22. Write a program using structures to add two distances in meter-kilometer format.
23. Write a program to add two complex numbers using structures.
24. Write a program to calculate the difference between two time periods using structures.